
Twitter or WhatsApp?

Public and Private Agent Networks Are Different at the Edge

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Agent networks are being built to look like Twitter or like WhatsApp, and the
2 two are compared as though “public versus private” were a variable. It is not. An
3 edge is a triple—who may form it, what it carries once formed, what is merely
4 asserted about it—and the regimes are two bundles that differ on all three at once.
5 Comparing them measures a bundle swap, and which number it returns is set
6 by a pairing nobody documents: the two diagonals of the same factorial give
7 0.181 and 0.314 on the same episodes. We give an instrument that sets the three
8 axes independently—one model, one runtime, one coordination kernel, conditions
9 expressed as capability grants—and a metric set that prices each axis in accuracy,
10 tokens and disclosure rather than in accuracy alone. Across 582 episodes of
11 hidden-profile coordination scored against forced-value answer keys, the factorial
12 recovers what neither diagonal isolates: 0.248, 0.066 and 0 on requirement F_1 . A
13 length-matched placebo moves the score as much as a trust assertion does, and
14 the formation effect is reach, vanishing when the roster already holds what a task
15 needs. On every axis we measured the bounded arm is weakly dominated—it
16 never scores higher and always costs $\sim 19k$ more tokens a task, including where it
17 merely ties—and two of its results were not designed. It is shown a quarter as much
18 misinformation as an open directory and writes more of it into the deliverable,
19 fabricating values 58 times where the open arms fabricated once: filtering out bad
20 claims also removes the second source that catches them. And a single brokered hop
21 recovers $+0.140$ of the gap while every assertion of trust remains flat, separating
22 an introduction, a formation intervention, from an endorsement, an attribute. If a
23 private agent network is worth building, the reason is not information quality, and
24 our measurements do not contain it.

1 Introduction

26 Two pictures of an agent network are in circulation. In the first, agents discover each other in the
27 open: they publish what they can do, broadcast what they need, and are matched by relevance rather
28 than acquaintance. In the second, agents act for people who already work together: membership is
29 bounded, interaction repeats, context accumulates. The first is called public and the second private,
30 and the difference is usually described as one of visibility.

31 Visibility is the wrong frame, and so is treating the two as endpoints of one dial. On Twitter an
32 account addresses any other, relevance is a claim in a profile, and a bad actor changes handles. On
33 WhatsApp a conversation exists because a number was already held or someone made an introduction,
34 the tie carries files and history, and misbehaviour costs the tie and embarrasses whoever vouched.

35 The two differ in what it costs to open an edge, in what the edge carries once open, and in what is
36 asserted about the counterpart—three things at once, not one.

37 Take the edge as the primitive and the three separate. **Edge formation** is which edges a requester
38 may create at all. **Edge semantics** is what an existing edge carries: a delegated task returning a result,
39 or direct access to state. **Edge attributes** are the beliefs attached to an edge—a reputation score, a
40 stated shared objective, a record that the parties have worked together before—which change nothing
41 a runtime checks. Public and private are two *bundles* in this space, not two definitions and not two
42 settings of one dial, so a study that compares them swaps all three at once. Worse, the two ways of
43 pairing the axes give different answers on the same data—0.181 and 0.314—so the reported size of
44 the “public advantage” is set by a bundling choice, not by a measurement. **Public versus private is**
45 **not the experimental variable**; $do(F)$, $do(S)$ and $do(A)$ are.

46 **This paper unbundles them and prices the components.** We place the two regimes as corners of
47 (F, S, A) rather than as definitions (§2). We give an instrument that sets the three axes independently,
48 and a metric set that charges each one in accuracy, tokens and disclosure (§3). And we state the design
49 question the decomposition makes answerable: not which regime wins, but which configuration a
50 setting should hold, and whether the axis weights are constants of the framework or of the task (§4).
51 The coordination kernel is an instrument, not a contribution.

52 **Contributions.** (i) **Public and private recast as two bundles in an edge space** $e = (F, S, A)$
53 rather than as two definitions, which makes “public versus private” a bundle swap rather than a
54 variable (§2). (ii) **An instrument that sets the three axes independently**—one model, one runtime,
55 one kernel, conditions as capability grants—with a metric set that prices an axis in accuracy, tokens
56 and disclosure rather than accuracy alone (§3). (iii) **A configuration problem in place of a binary**
57 **comparison**, whose first result is that the space is smaller than it looks and whose sharpest untested
58 prediction is that an edge’s position moves (§4). (iv) **A first decomposition on one task family**, with
59 a length-matched placebo on the attribute axis and two undesigned results (§5).

60 We do not study how a stranger becomes a teammate, and we make no safety claim about either
61 regime. We state in advance that the *existence* of a task-dependent crossing between regimes follows
62 from how we allocate facts and is not a discovery; what is measured is where it falls and what it
63 costs.

64 2 Public and private are two corners, not two definitions

65 Take the network as the unit of analysis: agents are nodes, a request from one to another is an edge.
66 An edge is not one thing. It is a triple

$$e = (F, S, A)$$

67 — **Formation**, who may open it; **Semantics**, what it carries once open; **Attributes**, what is asserted
68 about it. The three are independently setttable, and the first two are checked by a coordination layer
69 on every call while the third only ever reaches a model as text.

70 **Public and private are not points on one dial, and they are not definitions we**
71 **supply. They are two characteristic bundles in this space** — two corners that
72 co-occur in deployed systems because each component makes the others affordable.
73 Every real network sits somewhere between them, and most sit at no corner at all.

74 Twitter and WhatsApp are the two corners. On Twitter an account addresses any other without
75 assent, relevance is a claim in a profile, exchanges reveal nothing but their content, and trust arrives
76 as a platform-computed badge; a bad actor changes handles. On WhatsApp a conversation exists
77 because a number was already held or someone made an introduction, the tie carries files and history,
78 what is known about the counterpart is who introduced them and how the last exchange went, and
79 misbehaviour costs the tie and embarrasses whoever vouched.

80 The consequence is the paper’s premise:

81 **“Public versus private” is not an experimental variable.** It is a comparison
82 between two bundles that differ on three axes at once. A study that runs it measures

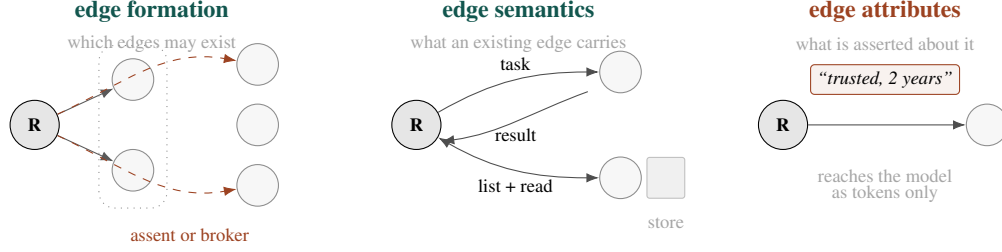


Figure 1: The three axes, drawn on $G = (V, E)$. **Formation** and **semantics** are enforced: a coordination layer refuses an edge that may not form and a request an edge does not carry. **Attributes** are never checked by anything—they arrive in a prompt, which is why asking whether a model acts on them is a real question rather than a tautology. Each regime supplies a characteristic value on all three (Table 1), and a study that compares public with private moves all three at once.

Table 1: The three axes, and what each regime supplies on it. Formation and semantics are enforced by a coordination layer on every call; attributes only ever reach a model as text. A study that compares a public network to a private one moves all three.

	Public	Private
Edge formation	Unilateral. Anyone in the directory is addressable; matching is by declared relevance. Lateral topology : wide and shallow, many weak ties.	Bilateral or brokered. A counterpart is reachable because of a prior tie or an introduction. Longitudinal topology : narrow and deep, few strong ties.
Edge semantics	Sandboxed by default. A counterpart accepts a task and returns a result; its store stays opaque. Interaction is one-shot request/response.	Openable. Shared workspaces, persistent namespaces, a store that may be listed and read—affordable precisely because the tie exists.
Edge attributes	Platform-computed for strangers: reputation scores, ratings, usage counts, verification badges.	Carried by the relationship: tenure, who introduced whom, how the last collaboration went, who is answerable.

83 $(F_{\text{pub}}, S_{\text{pub}}, A_{\text{pub}}) \rightarrow (F_{\text{pri}}, S_{\text{pri}}, A_{\text{pri}})$ and reports one number, which cannot be
84 attributed to any component. The variables are $do(F)$, $do(S)$ and $do(A)$, one at a
85 time.

86 Table 1 is what each corner supplies on each axis — the bundles unpacked.

87 Figure 1 draws the three on the graph. The rest of this section takes them one at a time.

88 **Edge formation** is which edges a requester may create: the scope of the directory it can enumerate
89 and address. This is the axis the word “public” usually refers to, and the one broadcast networks
90 for agents are built around—an agent publishes what it needs and what it offers, and the network
91 routes to whoever is relevant [EigenFlux]. Which links form, and at what price, is strategic network
92 formation [Jackson and Wolinsky, 1996]; the agent setting differs in that the price is conventionally
93 zero.

94 The two topologies are the substance of the axis. A public directory buys *breadth*: a rare requirement
95 finds someone, and—less obviously—a claim can be put to a second source. A private roster buys
96 *depth*: fewer counterparts, each already paid for, each with a history. Breadth and depth are not two
97 settings of one dial; they are the returns on two different edge prices.

98 **Edge semantics** is what an existing edge carries. A counterpart may accept a task and return a
99 result while revealing nothing else, or expose a store to be listed and read. The distinction is not ours:
100 it separates agent-to-agent task protocols, which deliberately keep a counterpart opaque [Google],
101 from granted resource scopes, in which one principal hands another read access to files. It maps
102 onto the regimes because exposure follows the tie—one does not open a working directory to a
103 stranger—but it is separately settable, and a capability kernel expresses both as grants over different
104 namespaces, so a condition is a set of capabilities rather than a different codebase.

105 **Edge attributes** is what is asserted *about* an edge. Both regimes have a characteristic form of
106 it, and both are trust-conveying metadata that no runtime checks: a public network computes a
107 score about a stranger, a private one carries a history about an acquaintance. They rest on the same
108 assumption—that an agent builds a representation of its counterpart from what it is told, and acts
109 differently because of it.

110 2.1 Why these three and not nine

111 The literature on ties, teams and open innovation supplies at least nine relational dimensions—
112 membership openness, identity continuity, shared-state depth, repeated interaction, relationship
113 origin, prior trust, shared goals, accountability, exit cost. They are not nine degrees of freedom: in a
114 deployed system an open directory brings weak identity, which brings diffuse accountability, which
115 brings low exit cost. The three-way grouping breaks that coupling the one way a controlled study can.
116 Formation and semantics are properties of the interaction layer—independently settable, checked
117 on every call, not open to argument by a model. Attributes are properties of the people behind the
118 agents; a harness cannot instantiate them, only *assert* them. That asymmetry is the experiment rather
119 than a limitation to apologise for: deployed networks communicate trust, alignment, attribution and
120 tenure as text on cards, so asking whether a model acts on that text asks whether a large part of
121 agent-network design does anything at all.

122 2.2 Two corrections the framework has to carry

123 Both fall out of taking the edge, rather than visibility, as the primitive.

124 **A public network’s noise is partly a feature.** The usual complaint is that an open directory is
125 full of unvetted claims. It is—but a requester who can address many counterparts can put the same
126 question to a second one, and contradiction is *detectable*. A roster that filters misinformation out
127 before it arrives also removes the second source needed to catch what does arrive. Whether the filter
128 or the redundancy dominates is an empirical question, not a definitional one, and the framework must
129 not settle it by calling breadth a cost.

130 **An endorsement is not an introduction.** These are on different axes and are routinely conflated.
131 “*This counterpart is reliable*” is an attribute: text, on an edge that already exists. *An introduction that*
132 *makes a third agent addressable* is formation: it changes the graph. A framework that files both under
133 “trust” predicts that a private network’s advantage comes from what it knows about its members. The
134 decomposition predicts something narrower—that only the half which changes reachability can do
135 work—and §5 reports which way it went.

136 3 Making each axis measurable

137 The framework is only useful if each axis can be set independently and each regime’s claimed
138 advantage can be priced. Model, runtime, task, answer key and coordination kernel are held fixed;
139 a condition is a set of capability grants rather than a different codebase, so *formation* (which cards
140 a requester may enumerate and address), *semantics* (whether a counterpart returns an answer or
141 exposes a store) and *attributes* (what the cards say about themselves and each other) move one at a
142 time. Formation and semantics are refused by the kernel when violated; attributes reach the model
143 only as tokens, which is what makes the third axis a test rather than a tautology. Appendix B gives
144 the instrument.

145 The task class has to force pooling, or reach cannot be worth anything. Each episode is a hidden-
146 profile deliverable: the facts a correct artifact needs are distributed across counterparts so that no
147 single agent, including the requester, holds enough [Stasser and Titus, 1985]. E , the share of required
148 facts placed *outside* the roster, is the dose on the formation axis—at $E = 0$ a private network needs
149 nothing it cannot already reach, so the axis must show no effect there. Distractors plant contradicting
150 values on counterparts other than the true holder, so that a second source has something to catch.
151 Every fact is a number, a date, a name or an enum, and the deliverable is scored by exact match after
152 a normalization rule fixed before the keys were written: no judge sits on any primary metric.

Table 2: Three settings, and what each predicts for the weight on each axis. These are predictions the framework makes and does not yet have evidence for; the task families are specified in Appendix A.8.

Setting	What a task looks like	formation	semantics	attributes
Enterprise team	Co-edit one artifact; the people are known; the context is the bottleneck	low	high	low
Personal errand	Find the one counterpart who can do a specific thing; no relationship needed after	high	low	medium
Personal social	Reach a person through someone; the introduction is the product	high, bro-kered	low	high

An axis is priced, not just scored. Around requirement F_1 sit useful sources (counterparts contacted that held a needed fact—the mechanism variable for formation), pollution absorbed and invented (a distractor’s value appearing in the artifact, split by whether the agent was ever shown it, the second being fabrication), and tokens and contacts—the price of the arm and, since each contact discloses the goal, its exposure. **Semantics** buys accuracy and is charged in tokens; reporting only the first half would make the private regime look better than the exchange rate supports.

Delivery is reported before any contrast. A trial that produced no artifact is a missing observation, not a score of zero, and averaging it in as zero converts an outage into an effect. We found this the expensive way (§A.7). Every contrast we report is therefore preceded by a per-cell delivery table, is computed both conditional on delivery and with losses scored as zero, and is treated as uninterpretable when delivery differs across compared cells by more than 15 points.

4 From a binary comparison to a configuration problem

Once the bundle is unpacked, the question changes shape. It is no longer which corner wins; it is which point in (F, S, A) a given setting should hold. Three binary axes give eight configurations, and no deployed network is a corner of them: a roster with an introduction protocol is public on formation for whoever a broker can reach and private for everyone else; an enterprise workspace is private on formation and fully open on semantics.

The first result of the decomposition is that the space is smaller than it looks. A configuration search is only a search if more than one dimension moves the outcome. On the task family we ran, the three weights are 0.248, 0.066 and 0 on requirement F_1 (§5), and the third is not small—it is indistinguishable from two neutral lines of text. **An axis with zero weight is not a dimension.** Eight configurations collapse to four. Semantics survives—0.066, interval $[+0.012, +0.119]$ —but it is charged at 11k tokens an episode, so on this task family the four collapse in practice to a choice on formation with a priced option attached.

That collapse is a finding, not a failure of the framework, and it is the kind of finding only a decomposition can produce: a study that compared the two corners would have reported one diagonal—0.181 or 0.314, depending on which pairing it happened to call public and private—and had no way to learn that one third of the bundle does nothing. We report it rather than restoring the eight-way search, because the honest next question is sharper than an optimisation. **Is the space two-dimensional everywhere, or only here?**

Scenarios are the test of whether the weights are constants. The collapse above was measured on a single task class: facts held by different agents, aggregated into one deliverable. That class is maximally favourable to formation—reach is literally what it asks for—and maximally unfavourable to semantics, since there is no shared artifact to co-edit. If the three weights are properties of the framework, they should hold elsewhere. If they are properties of the task, they will move, and the combination problem becomes real.

The enterprise row is the sharpest test, because it predicts the reverse of what we measured: when everyone needed is already reachable, formation has no work left to do, and the shared store that

cost 11k tokens for a small gain in a pooling task should pay for itself when the deliverable is edited jointly. It is also the row we defer—it needs a co-edited task family rather than a reallocation of ours, and we are pursuing it as dedicated work at a venue that can give the setting the space it needs. A framework that only ever produces “formation dominates” is a restatement of the task; one that predicts where formation stops dominating, and is then held to it, is a framework.

4.1 Edges may start public and end private

The framework’s sharpest prediction is not about a static configuration at all. If the three axes are separable, an edge’s position in (F, S, A) can *move*, and the direction is not arbitrary: each axis unlocks the next.

open discovery → sandboxed first exchange → repeated success
→ a persistent tie → attributes worth carrying → shared state

Read against our results this is not speculation about human networks. Public formation is what supplies a second source, and a second source is what our data says catches a wrong claim. Sandboxed semantics is what makes a first exchange with a stranger affordable. Repeated success is the only thing that could make an attribute mean anything—a reputation mark is inert here precisely because nothing generated it. And shared state is priced at 11k tokens, which is affordable only once an edge is expected to be used again. Each stage pays for the next.

This is the framework’s most testable prediction and we have not tested it. Every episode we ran is a cold start: 582 episodes at $\text{repeats} = 1$. The harness has the knob—a run may repeat a goal family instead of reworking one artifact—and it has never been turned. Two things would falsify the prediction cheaply. If an open requester’s contact set does not narrow across repeats, edges do not become private on their own and the transition must be designed rather than emergent. And if the attribute axis stays flat even when the assertions are *earned* rather than stipulated, then $w_A \approx 0$ is a fact about text and not about cold starts.

If instead the contact set does narrow, then “public versus private” is a question about *time* rather than architecture, and a private network is what a public one becomes.

What an optimum means here. An optimum is a configuration, not a size. On formation it is a *funnel depth*: how far past the roster a requester should be allowed to reach—own roster, one brokered hop, two hops, open directory—given that each additional layer widens reach and weakens recourse. The marginal return on a layer is the number of *independent* sources it adds, which saturates; the marginal cost is what a claim from that layer is worth, which decays. The crossing is the depth. Stated this way the optimum is measurable per scenario rather than argued per regime.

5 Evidence so far

We have run the decomposition on one task family: 582 episodes across sixteen sweeps, one model family, one kernel. The full inventory, every result including the nulls and one retraction, is Appendix A. Five findings bear on the framework.

Only formation is an intervention. A naive public-versus-private contrast returns 0.181 or 0.314 on requirement F_1 depending on which pairing is called public and which private—the two diagonals of one factorial, on one set of episodes. The factorial recovers what neither diagonal isolates: 0.248 (CI [+0.194, +0.302]) from edge formation, 0.066 (CI [+0.012, +0.119], and 19k tokens a task) from edge semantics, and nothing distinguishable from zero on attributes. That axis was tested across 56 contrasts per beat against a chance floor of 2.8; two cleared and reversed sign between beats of the same episodes, and a length-matched placebo of two neutral lines moved the score by +0.005. Prior trust and attribution, the two properties deployed reputation systems carry, are the flattest measurements in the set.

The formation effect is reach, and the dose-response says so. With E —the share of required facts outside the roster—as the dose, the contrast runs +0.034 (95% CI [−0.067, +0.137], spans zero) at $E = 0$ (Figure 2), +0.111 at $E = 0.3$ and +0.279 at $E = 0.7$. The zero-dose cell is the control the

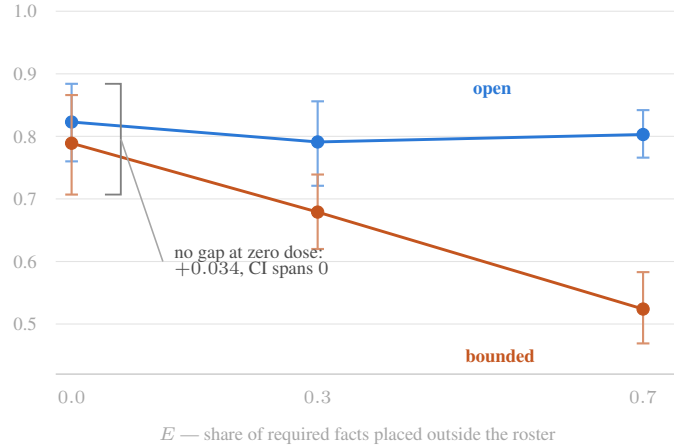


Figure 2: The formation effect is reach, and the dose-response says so. E is the share of a task’s required facts placed outside the roster; bars are 95% percentile bootstrap intervals over episodes. Four things hold at once, and the axis needs all four: the contrast **vanishes at zero dose**, which is the control it had to pass; the arm that cannot be affected is **flat** across all three doses; the mechanism variable tracks the dose (the bounded arm’s useful sources fall $4.36 \rightarrow 3.17 \rightarrow 1.26$); and fabrication appears only as sources dry up ($0 \rightarrow 0.139 \rightarrow 0.333$ per task).

axis had to pass and did: a private network that can already reach what a task needs is not worse. The public arm is flat across all three doses (0.823/0.791/0.803), so the manipulation is specific to the arm it should affect, and the mechanism variable tracks it—the private arm’s useful sources fall $4.36 \rightarrow 3.17 \rightarrow 1.26$.

A roster sees less bad information and believes more of it. This is the one result we did not design (Figure 4, in the appendix). The private arm is shown roughly a quarter as much planted misinformation as the public arm, and writes it into the deliverable *more*—0.269 against 0.094 instances per task (difference $+0.175$, CI $[+0.105, +0.242]$). When it reaches no one at all it does not decline to answer: across matched sets of 287 and 286 delivered tasks, the public arms fabricated a value once and the private arms 58 times. Filtering misinformation out before it arrives also removes the second source that would have caught it, which is the redundancy argument of §2 appearing as a failure mode rather than as a benefit. That explanation is not the whole mechanism, and we say so because the tidier version is available and wrong: among beats that obtained four useful sources the absorption rate is 2.4% open against 18.1% bounded, so starvation amplifies credulity without being all of it.

An introduction does work an endorsement does not. A bounded roster given a single brokered hop—one roster member may forward a request to a counterpart the requester cannot address—recovers close to half the gap to the open arm, and most of the fabrication with it; letting a bounded arm *read* the whole directory while keeping deep relationships inside the roster is indistinguishable from full openness (Figure 3). Both are formation manipulations, and both move. Every attribute manipulation is flat, including the ones asserting exactly the trust an introduction would carry. This is the endorsement/introduction distinction of §2 landing where the framework said it would: what changes the graph does work, what describes it does not.

On everything we measured, the bounded arm is weakly dominated. The decomposition invites a symmetric reading that the data does not support, so this is worth stating without hedging.

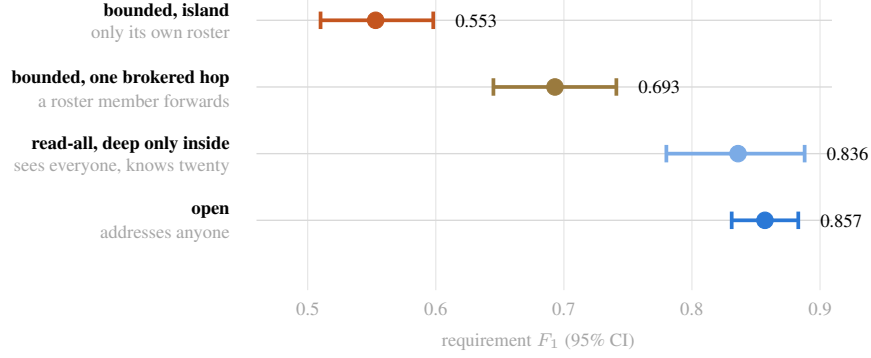


Figure 3: An introduction does work an endorsement does not. All four rows are formation manipulations, and every one of them moves; every **attribute** manipulation—including the ones asserting exactly the trust a broker would carry—is flat. One brokered hop recovers +0.140 (CI [+0.073, +0.204]) of the 0.304 gap and cuts fabrication from 0.223 to 0.057 per task; letting a bounded arm read the whole directory while keeping deep relationships inside the roster is indistinguishable from full openness (+0.019, spans zero). Depth of relationship is not what is bought here—a second reachable source is.

bounded – open	$E = 0$	$E = 0.3$	$E = 0.7$
requirement F_1	tie	−0.111*	−0.279*
tokens per task	+18.8k*	+19.5k*	+19.1k*
wall clock	tie	+0.8 min*	+0.5 min*
fabrications per task	tie (both 0)	+0.139*	+0.319*
counterparts disclosed to	tie	tie	−1.7*

* 95% CI excludes zero.

The $E = 0$ column is the private network’s best case: its roster already holds every fact the task needs. It does not win there. It ties on accuracy and spends 38% more tokens to do it—an overhead of +17.9k in one harness generation and +19.3k in the other, so not an artifact of either. Its growth is explained (the bounded arm searches its twenty-card roster $6.8 \rightarrow 10.1 \rightarrow 16.2$ times as E rises while its useful sources fall, which is what looking harder for something that is not there costs); the baseline at $E = 0$ is not. The one column where it looks better is disclosure, and only at $E = 0.7$, where it also scores 0.279 lower and fabricates in a third of its tasks. **A network that leaks less because it reached nobody has not protected anything.**

So if a private agent network is worth building, our measurements do not contain the reason. Three candidates sit outside the scorer, and each names its own experiment. *Confidentiality*: our only proxy is how many counterparts learn the goal, and it favours the bounded arm only where that arm has failed—a real test needs a disclosure term in the objective, not a by-product of contact count. *Recourse*: accountability needs a second encounter to mean anything, and every episode here is a cold start (§4.1). *Adversaries*: 36 of 100 cards carry plausible tags and no knowledge, but every agent that is asked answers honestly, so the defence a private network exists to provide has nothing to defend against here. What the data does rule out is the justification most often given: that a bounded network produces better information. It produces worse information, more expensively.

Two further limits are ours rather than the framework’s. One model family carries every sweep with a live main effect—the second-model replication was run only at $E = 0$, the control cell, so model-independence is not established. And the scenario predictions of §4 are predictions.

6 Further question: the two regimes scale on different bottlenecks

The framework invites a scaling question it does not answer, and unpacking the bundle changes what the question is. The two corners do not grow along the same variable, so they cannot degrade for the same reason.

290 A public network grows in N , the number of addressable counterparts, and N is not chosen by
 291 anyone. Coverage rises with it; so do search cost, verification cost and exposure to counterparts
 292 nobody vetted. Its binding constraint is whether ranking can still surface the right counterpart out of
 293 N .

294 A private network cannot grow in N : a roster is bounded by construction. It grows in d , the number
 295 of brokered hops a request may travel, because that is the only way trusted coverage increases. Its
 296 costs are not retrieval costs—they are the cost of asking a broker to spend its own standing, and the
 297 decay of whatever recourse survives a chain of introductions. Its binding constraint is path length and
 298 trust transitivity.

299 **Public networks scale through ranking; private networks scale through rela-**
 300 **tionships.** The open question the framework poses is not which is better at scale but
 301 which degrades faster, under what task distribution, and whether a hybrid inherits
 302 the better bottleneck from each.

303 We can speak to the first of the two and not the second, and only within a range. Growing the directory
 304 from 50 to 800 cards— $16\times$, roster held at twenty—leaves the open arm’s score flat ($0.800 \rightarrow 0.826$,
 305 CI spans zero) while search precision falls by roughly half and tokens double: the arm pays for the
 306 extra difficulty rather than failing at it. We report this null with a caveat earned the hard way, since
 307 our first reading of the same sweep was a scaling result and was an outage (§A.7).

308 Extrapolating past that range needs a model of how well a directory can be ranked, and the honest
 309 version is uncomfortable. A card is ~ 50 words of self-description, so what ranking must recover—
 310 whether this agent holds *this* value—is a property of private state and is not in the index. Writing ρ for
 311 the correlation between rank and holding the fact, $\rho = 0$ is not the honest case but the *no-signal* case;
 312 web search retrieves from 10^{11} documents, so ranking plainly can be good. The question is how good,
 313 and under the model of §A.9 (Figure 6) the answer is that $\rho = 0.9$ has already collapsed by 2×10^5
 314 cards and reaching 10^6 needs $\rho \approx 0.999$. “A better algorithm” understates it: web search reaches its
 315 ρ by aggregating outcome feedback rather than by sorting more cleverly, and the equivalent signal
 316 in an agent network is a record of who has answered correctly before—which is what a reputation
 317 mechanism would have to bootstrap, and our one test of that was null for a reason we can name
 318 (§A.6).

319 **The private side is one data point.** The brokered hop of §5 is $d = 1$. That single hop recovers
 320 $+0.140$ of a 0.304 gap; whether $d = 2$ adds another increment, half of one, or nothing is the shape
 321 of the private curve and we have not measured a second point on it. The framework predicts the
 322 increments shrink faster than reach grows, because what is spent at each hop—a broker’s willingness
 323 to attach its own standing to a stranger—is exactly what attenuates with distance while reachable
 324 counterparts multiply. If that holds, a private network has an interior optimum in d where a public
 325 one has none in N , and the two curves are not two settings of one model.

326 7 Related work

327 Agent-to-agent task protocols publish a capability card and accept tasks while deliberately keeping
 328 a counterpart opaque [Google]; broadcast networks route published needs among strangers while
 329 stating that private information does not enter the network [EigenFlux]; surveys catalogue the wider
 330 design space [Yang et al., 2025, Georgio et al., 2025]. These encode exactly the split we intervene on
 331 — whether an edge carries a delegated task or direct access to state — and capability discovery is the
 332 first thrust of current agentic-web research, but to our knowledge neither has been evaluated for which
 333 side of it changes collective behavior. Which links form, and at what cost, is the subject of strategic
 334 network formation [Jackson and Wolinsky, 1996]; our formation axis is that question with the cost
 335 set to zero, which is the assumption a follow-up should relax first. The intuition that open networks
 336 supply variety and bounded ones supply follow-through descends from weak ties [Granovetter, 1973],
 337 exploration and exploitation [March, 1991], transactive memory [Wegner, 1987], and open innovation
 338 [Chesbrough, 2003], all studying humans, where the properties in our attribute group are real rather
 339 than asserted. Our tasks are the hidden-profile paradigm [Stasser and Titus, 1985], which supplies
 340 exact ground truth and makes pooling rather than retrieval the object of measurement; the authority
 341 model is object-capability discipline [Miller, 2006], so a condition is a set of grants a kernel checks
 342 rather than a convention an agent could ignore. Where an answer key cannot reach, blind pairwise

comparison with a cross-family judge is the standard remedy [Zheng et al., 2023], since evaluators favour their own generations [Panickssery et al., 2024]; we preregistered such a protocol but did not run it, and no result here depends on a judge.

8 Conclusion

Public and private agent networks are two bundles of edge properties, not two values of one variable. An edge is a triple—who may form it, what it carries, what is asserted about it—and Twitter and WhatsApp are two corners of that space, not its two poles. Comparing the corners moves all three axes at once, which is why the comparison has never told anyone which component was doing the work, and why it returns 0.181 or 0.314 on the same episodes depending on which pairing an author happens to call public and private.

Separated on one kernel, one runtime and one model, the three weigh 0.248, 0.066 and 0 on a hidden-profile task family. The formation advantage is reach: it vanishes when the roster already holds what a task needs. The semantics advantage is real, small, and charged at 19k tokens a task. The attribute axis does not move, and a length-matched placebo of two neutral lines is indistinguishable from an assertion of prior trust. **The first result of a decomposition is that the space is smaller than it looks:** an axis whose effect cannot be told apart from two lines of filler is not a dimension.

On this task family the bounded network is weakly dominated, and we do not want to be read as splitting the difference. It never scores higher. It costs $\sim 19k$ more tokens a task at every E we ran—including $E = 0$, where its roster holds everything the task needs and it merely ties. It is slower. It fabricates values 58 times where the open arms fabricated once, because the filter that keeps bad claims out also removes the second source that catches them. Its single better column, disclosing to fewer counterparts, appears only where it has already failed to reach anyone, which is not a privacy property. Against that, one brokered hop recovers close to half the gap while every assertion *about* trust stays flat: an introduction changes the graph, an endorsement does not.

So if a private agent network is worth building, our measurements do not contain the reason, and the reason is not information quality. Confidentiality, recourse over repeated encounters, and resistance to counterparts with an incentive to lie are the three candidates, and all three sit outside our scorer—each naming an experiment rather than a hedge. Two further questions are the framework’s own: whether the space is two-dimensional everywhere or only on a task family that asks for reach, and whether an edge’s position is fixed at all. Open discovery buys the second source, a sandboxed first exchange makes a stranger affordable, repeated success is the only thing that could make an attribute mean anything, and shared state is worth its price only once an edge expects to be reused. If contact sets narrow across repeats, a private network is what a public one becomes, and the regimes were never alternatives. For an agentic web the implication is narrow and, we think, useful: changing what a network permits changes what the collective does, and describing the relationship does not—not at the layer where descriptions currently live.

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413 A Everything that was run

414 Sixteen sweeps, 582 episodes, 2,328 beats, one model family except where noted. This appendix is
415 the complete record: what each sweep varied, what it establishes, and the two that establish nothing.
416 Numbers are recomputed from per-episode summaries over delivered beats, with percentile bootstrap
417 intervals; §3 states why delivery is tabulated before any contrast.

Table 3: Every sweep. “ep” is episodes; each episode is four beats.

Sweep	Varied	ep	What it establishes
realistic	$2 \times 2 \times E$	72	Main decomposition: $0.248/0.066/\approx 0$
axis2x2	$2 \times 2 \times E$	72	Independent replication of the same
matched-E0	three arms, $E = 0$	27	Zero-dose control: the gap disappears
main	three arms $\times E$	54	Read-all $\approx$ open; public – private = +0.240
matched-E0-glm	second model, $E = 0$	18	The control replicates; the main effect is untested
scaling	edge cost $\times$ relay depth	112	Brokered hop +0.140; per-edge cost is null
seedaxis	six attribute profiles	72	Attribute axis flat; placebo +0.005
repscan	reputation off/real/random	48	Null, but the design is at fault (§A.6)
capscan	search cap 5/20/80	24	Reading $16\times$ more results buys nothing
nscan	directory 50–800	60	Retracted (§A.7)
k-pilot	deliverable complexity	8	Pilot only
five smoke sweeps	mechanism checks	15	Confirm each manipulation fires

418 A.1 How the headline cells are computed

419 Every number in the 2×2 (Figure 7) is the mean over *delivered* cold-beat episodes of *realistic*
420 and *axis2x2* pooled—the two runs share a design, and pooling doubles the cells to 35–36 episodes
421 each. Intervals are percentile bootstrap over episodes, resampled jointly across cells so that a contrast
422 between two means is resampled as a contrast.

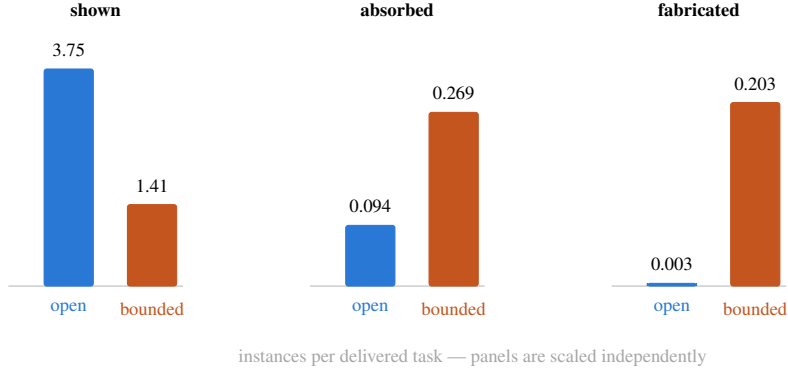


Figure 4: The result nobody designed. A bounded roster is shown roughly a third as much planted misinformation as an open directory—the filter works—and writes almost three times as much of it into the deliverable (+0.175, CI [+0.105, +0.242]). Worse, when it reaches no one it does not decline to answer: across 287 and 286 delivered tasks the open arms fabricated a value *once* and the bounded arms 58 times. Filtering bad claims out before they arrive removes the second source that would have caught them, so the redundancy argument of §2 shows up here as a failure mode rather than as a benefit.

423 An earlier draft reported this table from `axis2x2` alone with lost beats scored as $F_1 = 0$. One lost
 424 cold beat in cell *A* moved that cell by 0.046 and both diagonals with it, which is the failure mode §3
 425 warns about appearing in our own headline table. The values here are the corrected ones.

426 A.2 Formation: the dose-response and its control

427 *E* is the share of required facts placed outside the roster.

<i>E</i>	open	bounded	Δ	95% CI		bounded sources	bounded fabrications
0.0	0.823	0.789	+0.034	[−0.067, +0.137]	spans 0	4.36	0.000
0.3	0.791	0.679	+0.111	[+0.020, +0.200]		3.17	0.139
0.7	0.803	0.524	+0.279	[+0.210, +0.345]		1.26	0.333

430 Four properties hold together, and the paper’s use of this axis rests on all four: the zero-dose contrast
 431 spans zero; the arm that should be unaffected is flat (0.823/0.791/0.803); the mechanism variable
 432 tracks the dose; and the failure mode appears only as sources dry up. The $E = 0$ cells come from the
 433 three-arm generation of the harness (matched-E0), which is why they are reported alongside rather
 434 than inside the 2×2 .

435 A.3 The credulity result

436 Pooled over `realistic` and `axis2x2`, restricted to delivered beats (287 open, 286 bounded):

	open	bounded	difference
misinformation shown	$\sim 4\times$ more	baseline	the roster is genuinely cleaner
absorbed into the deliverable	0.094 / task	0.269 / task	+0.175 CI [+0.105, +0.242]
fabricated outright	1 instance	58 instances	—

438 `absorbed` requires that the agent was shown the distractor; `fabricated` requires that it was not, so
 439 the second is a value invented that happens to coincide with a planted wrong one—a conservative
 440 lower bound on fabrication, since only coincidences with planted values are detectable. Stratifying by
 441 the number of useful sources obtained does not remove the gap: at four useful sources the absorption
 442 rate is 2.4% open against 18.1% bounded, so starvation amplifies the effect but is not the whole of it.

443 A.4 Introduction against endorsement

444 From scaling, arm C, relay depth 0 against 1, balanced across all three edge-cost levels (28 episodes
445 each):

	Architecture	F_1	useful sources	absorption	fabrications/task
446	bounded, island	0.553	1.50	11.0%	0.223
	bounded, one brokered hop	0.693	3.40	6.3%	0.057
	read-all, deep only inside	0.836	—	—	0.000
	open	0.857	4.63	2.4%	0.000

447 The hop is worth +0.140 (CI [+0.073, +0.204]) conditional on delivery and +0.088 (CI
448 [+0.016, +0.158]) with lost beats scored as zero, and it clears zero independently within each
449 edge-cost stratum (+0.128, +0.138, +0.173). Read-all against open is +0.019 and spans zero. This
450 supersedes an earlier estimate of +0.042 (CI [-0.137, +0.208]) computed from a 16-episode cell;
451 the number in Table 3 is the 28-per-arm one.

452 A.5 Nulls worth reporting

453 **Per-edge cost does not move the ranking.** Charging a handshake turn for each contact outside the
454 roster—194 and 384 handshakes fired at costs of one and two turns—leaves the open arm’s advantage
455 at +0.219, +0.263, +0.244 across the three levels while its tokens rise from 53k to 74k. The arm
456 absorbs friction as cost rather than as failure. An earlier draft named this as the variable most likely
457 to reverse the ranking and stated it had not been run; both halves of that sentence are corrected here.

458 **Reading more search results buys nothing.** Caps of 5, 20 and 80 give 0.791, 0.755, 0.819, every
459 contrast spanning zero.

460 **Attributes.** Six profiles against a length-matched placebo; see §N for the seed texts and §5 for the
461 numbers.

462 A.6 Reputation: a null caused by the task, not the mechanism

463 Marks were differential—counterparts that held a planted falsehood were flagged, and a random-mark
464 arm held the wording fixed while removing the information. All three arms are identical: 0.821 off,
465 0.817 real, 0.833 random, every contrast spanning zero.

466 The design is at fault. Each fact has exactly one holder, so a flagged counterpart is also the only route
467 to the true values it holds; a requester that avoided flagged cards would lose more than it gained,
468 and routing measurement confirms it moved *toward* them. Making the mark actionable requires
469 two holders per fact, one reliable and one not, which is a change to fact allocation rather than to
470 the reputation mechanism. We report this as a failed manipulation rather than as evidence about
471 reputation.

472 A.7 The retracted directory-size sweep

473 nscan varied the directory from 50 to 800 cards and appeared to show the open arm’s advantage
474 decaying to zero. It does not. 27% of its beats produced no artifact, every one of them ending in a
475 transport failure after four retries, and the scorer’s $F_1 = 0$ for a beat with no artifact was averaged in
476 as a score. Other sweeps lost 2–3%.

477 Three checks establish that the losses are not caused by directory size. The bounded arm reads a
478 twenty-card roster at every setting, so size is not in its causal path, yet its delivery rate rose from
479 54% to 96% across the sweep (Figure 5). Scenario was collinear with wall clock—one scenario ran
480 entirely inside the failure window—and the two arms walked their size settings in opposite orders
481 inside it, which is what drew one line down and the other up. And lost beats used 4.4 of 22+ available
482 turns with zero build calls while taking *longer* in wall clock, which is a stalled connection rather than
483 an exhausted budget.

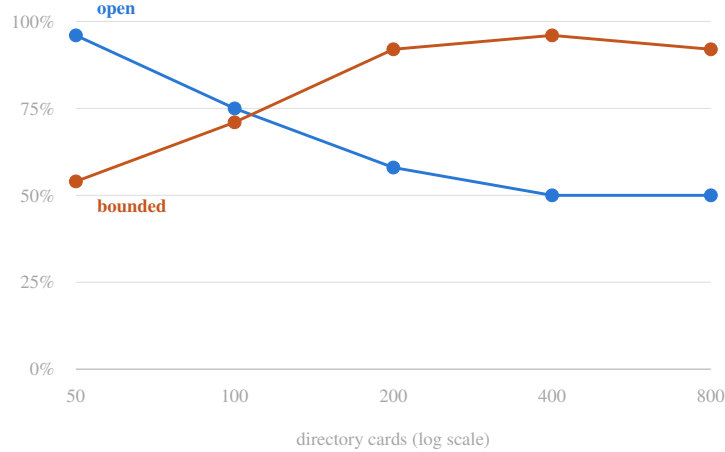


Figure 5: Delivery rate in the retracted sweep: the share of episodes that produced an artifact at all. The open arm falls with directory size and the bounded arm rises. The bounded arm reads a twenty-card roster at every setting, so directory size is not in its causal path and its delivery rate cannot be responding to it. Both curves are the same transport outage, sampled by two arms that happened to walk their size settings in opposite orders. Scoring the missing artifacts as $F_1 = 0$ turned this figure into a scaling result.

Conditional on delivery both arms are flat: the open arm runs 0.800, 0.777, 0.758, 0.779, 0.826 across a $16\times$ range ($\Delta = -0.026$, CI $[-0.123, +0.065]$). In the one cell with full delivery at every size, search precision falls from 0.528 to 0.386 while the score holds and tokens double—the finding reported in §6, at $n = 3$ episodes per point and therefore as a hypothesis for a rerun.

The analysis code now computes per-cell delivery before any contrast and refuses to interpret one when delivery differs across compared cells by more than 15 points.

A.8 Task families for the scenario test

The scenario predictions of §4 require three task families rather than one. Each keeps forced-value scoring and hidden-profile allocation; what changes is where the difficulty sits.

- **Enterprise team.** One artifact revised across four beats by counterparts who all hold part of the context and all remain reachable ($E = 0$). Formation is starved by construction; the shared store is the only route to the accumulated state. This is the row that predicts the reverse of what we measured.
- **Personal errand.** A single specific capability is needed once, held by one counterpart among many that claim adjacency, with no follow-up beat. Formation should dominate more sharply than in the pooling family, and semantics should be worthless.
- **Personal social.** The deliverable requires reaching a counterpart who is addressable only through a chain of brokers, with each hop’s willingness a function of what the previous hop asserts. This is the only family in which an attribute is load-bearing, because it gates whether the next edge forms at all—which, if it holds, would locate the attribute axis’s one real use.

A.9 The retrieval model behind the scaling question

Let k be the number of search results a requester acts on, m the number of cards matching a query, h the number of those that actually hold the needed fact, and ρ the correlation between a card’s rank and holding it. With no ranking signal the needed holder’s expected rank is $(m + 1)/2$; with perfect ranking it is $(h + 1)/2$. Interpolating between the two by ρ and truncating at k gives the survival probability plotted in Figure 6. We take $m(N) \approx 0.094N$ from the directory generator, which draws 35% of filler tags from the scenario vocabulary, and hold $k = 20$.

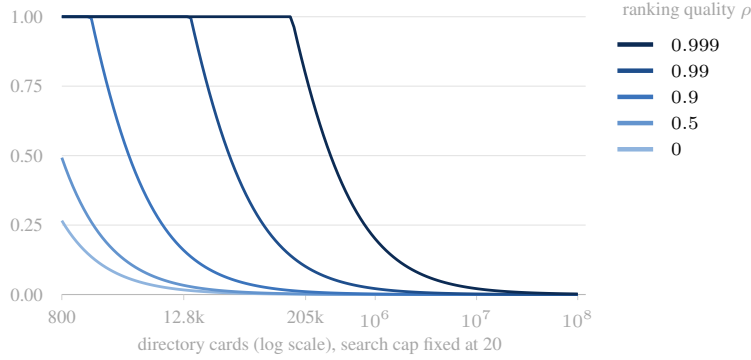


Figure 6: How good must ranking be for a public directory to keep working. ρ is the correlation between a card’s rank and whether its owner actually holds the needed fact; the vertical axis is the probability that the holder survives truncation at 20 results, under the model of §A.9 evaluated continuously in N . $\rho = 0$ is not the honest case but the *no-signal* case—web search retrieves from 10^{11} documents, so ranking plainly can be good. The question is how good, and the answer is uncomfortable: $\rho = 0.9$ has collapsed by 2×10^5 cards, and reaching 10^6 needs $\rho \approx 0.999$. “A better algorithm” understates it; web search reaches its ρ by aggregating outcome feedback, which is what a reputation mechanism would have to bootstrap (§A.6).

511 The model is deliberately crude—it assumes a single query and ignores reformulation—and its
 512 purpose is not to predict a number but to establish an order of magnitude for the requirement on ρ .
 513 That requirement is the load-bearing claim: at 10^6 cards the ranking has to be right to within one part
 514 in a thousand, which is not a threshold a better similarity function reaches.

515 B Interventions on the graph

516 Formation and semantics are interventions on different parts of $G = (V, E)$, and Figure 7 exists
 517 to keep them apart. **Formation** changes which edges a requester may create: open discovery lets it
 518 address any of 100 holders, bounded discovery restricts it to a roster of 20. **Semantics** changes what
 519 an existing edge carries: e_{ij} either delegates a task and returns a result, or grants `list` and `read over`
 520 a store.

521 Two further dimensions ride on the same mechanism and are set with the cell rather than varied
 522 independently: identity continuity is whether a counterpart’s namespace survives the encounter, and
 523 repeated interaction is whether later beats reuse it. Delegation depth is bounded per cell and refusals
 524 are logged, so a counterpart that tries to sub-delegate past its ceiling produces an event rather than a
 525 silent success. The kernel is an instrument: it lets us set properties deployed systems couple together,
 526 and it refuses an agent that tries to exceed them.

527 C Seeding the textual dimensions

528 Appendix N gives the seed text and the behavioural readout for each profile. The five attribute
 529 dimensions are properties of the people behind the agents, and a harness cannot instantiate them.
 530 It can assert them—which is exactly what a deployed agent card, reputation field, or relationship
 531 summary does. We therefore seed each as context and measure whether behavior changes.

532 **The origin condition is the cleanest instrument.** Four seeds change what is stated about incentives
 533 or consequences. **RELATIONSHIP ORIGIN** changes only how the encounter is narrated: same directory,
 534 same grants, same knowledge, same store. If the primary metrics move under that manipulation alone,
 535 relational framing does real work; if they do not, richer metadata on a card is unlikely to do better,
 536 because it is the same channel carrying more words.

537 **Validity boundary.** *Seeded trust is not trust.* We measure whether an agent acts differently when
 538 told it is in a trusting, aligned, accountable or long-standing relationship—the sensitivity of a runtime

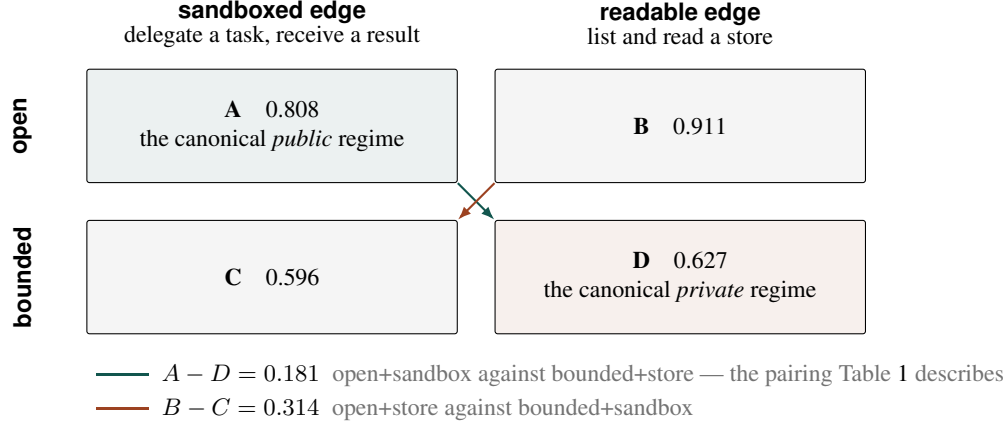


Figure 7: Requirement F_1 at the cold beat, over 35–36 delivered episodes a cell. **Both diagonals are honest “public versus private” comparisons, and they differ by $1.73\times$.** Which one a study reports depends on a bundling choice nobody documents: pair open discovery with a sandboxed edge and bounded discovery with a readable one—the pairing Table 1 describes—and the answer is 0.181; pair them the other way and it is 0.314. Neither is the effect of anything. The factorial recovers what is: $\Delta_{\text{formation}} = \frac{1}{2}[(A - C) + (B - D)] = 0.248$ and $\Delta_{\text{semantics}} = \frac{1}{2}[(B - A) + (D - C)] = 0.066$. The two diagonals are $\Delta_{\text{formation}} \pm \Delta_{\text{semantics}}$: on $B - C$ the axes reinforce and the number overstates formation by 27%, on $A - D$ they oppose and it understates it by the same 27%.

to asserted relational context, which is what an agent network actually delivers to a model. We do not measure what trust does in a human network, and no claim here should be read that way. If seeds move behavior the paper reports elasticities usable in card and protocol design; if not, it reports that relational metadata is inert without architectural backing.

D Tasks and measurement

D.1 Deliverables with forced values

Every task is a hidden-profile artifact. The facts a deliverable requires are split across specialist counterparts, and plausible-but-wrong *distractors* are planted alongside them—each on a counterpart other than the holder of the value it contradicts, and at least one inside the bounded roster, so no configuration is immune by construction. Ground truth is designed rather than inferred, so scoring is set arithmetic.

Two schema properties do the work. Every required field is *forced*: its value is determined by the facts, so no field can be a matter of taste. And one fact set has two projections—a flat *report* whose fields are values counterparts hold, and a *chained decision* whose fields depend on one another—which makes coordination depth a real dimension rather than a chain welded onto a flat task. Elimination forces pooling: one constraint rules out one option, a second another, a third a third, each on a different counterpart, so the surviving choice is unreachable from any one of them. Tasks also carry a *craft* component, a rendered artifact with structural requirements, because a value can be retrieved in one message while a working artifact cannot—which is where iteration and delegation depth begin to matter. Three scoring rules are fixed before any key is written (Appendix I): the schema and its enumeration vocabulary are given to the agent so only values are unknown, normalization is specified in advance, and evidence is matched on values rather than internal identifiers.

D.2 Metrics

Five metrics are exact-match or counters and carry the paper’s claims: `requirement_f1`, the F_1 over required values against distractors asserted; `completion`, schema conformance used as a gate; `realization`, achieved recall divided by the recall attainable given the configuration’s reach;

Table 4: Predictions, fixed before the runs, with the outcome. Two survived, one was falsified in a way that changed the paper’s second finding, and one was withdrawn.

	Prediction	Falsified if
H1	Edge formation dominates edge attributes	attribute contrasts move outcomes comparably to formation
H2	Attributes are inert once text length is controlled	a relational seed beats a length-matched placebo consistently
H3	Sandboxed edges beat readable ones on accuracy	readable edges are at least as accurate
H3'	Sandboxed edges win <i>per token</i>	readable edges are more efficient per token in either arm
H4	The formation advantage is a reach advantage, and shrinks when reach stops being scarce	the gap is flat in E and across beats
H5	Readable edges absorb more misinformation	absorption tracks exposure alone

boundary_cost, the sanitize, filter, verify and distil actions with their tokens; and crossing_count, how many times a hybrid episode changed regime.

realization is what keeps the study honest. The allocation fixes how much any configuration could obtain; dividing it out leaves what the configuration converted—whether it found the right counterparts, asked enough of them, and resolved the conflicts. The ceiling is designed; the realization is not. Two instrumentation rules follow from a defect we want to make impossible: completion is decided by the harness’s deterministic checker and never read from a condition-specific field—such a field exists on the private path of the deployed system we draw from and has no public counterpart, so using it would let “private completes more” prove itself—and token accounting is counted in the harness for the same reason.

Three quantities an answer key cannot reach—overall utility, novelty, and diversity of approach—were preregistered for blind pairwise scoring by a cross-family judge, with agreement reported. **They were not run for this submission.** No result in this paper depends on a judge, and we list them as planned work rather than describe a protocol we did not execute.

Honesty statement. The allocation fixes how much each configuration can reach, so the *existence* of a task-dependent crossing is a property of the design and not a discovery. What is measured is where it falls, what it costs, and how much of the attainable each configuration realizes. The seeded conditions and the access axis carry no such guarantee, and they are the study’s primary empirical content.

E Experimental design

E.1 Predictions

Each prediction was chosen against one filter: *would a careful reader bet on it in advance?* Claims that pass trivially—an open network reaches more knowledge, a persistent team completes more—are excluded, because confirming them consumes a study and returns nothing. Every prediction below was fixed before the corresponding sweep ran, together with the condition that would falsify it.

H3 is worth naming as a failure. We predicted that a counterpart answering from its own full context would beat a requester re-joining fragments it had read—that extraction is lossy in a way delegation is not. On accuracy it is not: readable edges are ahead in both arms. The prediction survives only once cost is counted, which is a different and weaker claim, and we report it as H3' rather than presenting the original as confirmed.

E.2 Conditions

Four cells of the 2×2 , each a configuration of the same kernel: open or bounded edge formation crossed with sandboxed or readable edge semantics. Cells B and C are what allow the two main effects to be identified rather than summed; without them, an open-versus-bounded contrast reports $\Delta_{\text{formation}} + \Delta_{\text{semantics}}$ and cannot say which term carried it.

Table 5: Requirement F_1 at the cold beat, by cell, and the two main effects. The naive regime contrast is 0.314 on one diagonal and 0.181 on the other; on the first, three quarters of it is who can be reached and one quarter is what crosses the edge.

	sandboxed	readable	
open	0.808	0.911	
bounded	0.596	0.627	
$\Delta_{\text{formation}} = 0.248$	95% CI [0.194, 0.302]		excludes 0
$\Delta_{\text{semantics}} = 0.066$	95% CI [0.012, 0.119]		excludes 0

The attribute axis holds the configuration fixed and varies only what is asserted in context, across five relational profiles and a length-matched placebo.

E.3 Controls

One model across all agents and all cells, one runtime, one kernel; identical task prompts, deliverable schemas, scoring functions, token budgets and seed distributions. Rate limits and loop bounds are logged per episode rather than absorbed into results. Rubrics, schemas, normalization rules and the scorer signature were frozen before any run (Appendix J).

Two controls deserve naming because each removed an explanation that would otherwise have survived into the paper. The **requester keeps its own notes in every cell**: withholding them from the open configurations would have manufactured the warm-start result, so what differs is not memory but whether memory is bilateral. And the **length-matched placebo** carries the same line count, position and register as the relational seeds with no relational content, which is what separates an effect of the relationship from an effect of the tokens describing it.

F Results

All numbers below come from 144 episodes on one kernel, one runtime, and one model: 72 for the 2×2 over edge formation and edge semantics, 72 for the attribute axis. Bootstrap intervals are percentile over episodes, resampled within cell.

F.1 A naive public-versus-private comparison, decomposed

The contrast a study of “public against private” would report is open+store against bounded+sandbox: 0.314 on requirement F_1 at the cold beat. It decomposes almost exactly additively into the two interventions.

F.2 Edge formation dominates (H1)

Across seven tests—four accuracy metrics and three cost metrics, over the cold, rework and warm beats— $\Delta_{\text{formation}}$ clears its interval in six. Chance at $\alpha = 0.05$ would produce 0.4. The sign is stable and the magnitude is largest where the task most needs outside knowledge: at a high external fraction it reaches 0.352, against 0.112 when most holders sit inside the roster.

It is also cheaper. Opening the directory saves 17.1k tokens per episode (CI excludes 0) while raising accuracy, because a requester that can address the right holder stops paying for the search around it.

F.3 Edge semantics: a small gain at a real price (H3)

$\Delta_{\text{semantics}}$ is positive at every beat—0.066 cold, 0.031 on rework, 0.084 warm, 0.055 on warm rework—and the cold and warm intervals exclude zero once the two runs are pooled. It is small, consistent, and, as the cost side below shows, expensive.

The cost side is unambiguous, and it inverts the ranking. Reading a counterpart’s store costs 11.0k tokens per episode (CI excludes 0). Per token spent, agent mediation is ahead in both arms:

Table 6: Attribute contrasts on requirement F_1 , stratified by (arm, scenario) over four strata, $n = 12$ per profile. The placebo matches the relational seeds in line count, position and register while carrying no relational content.

contrast	cold	warm	
placebo – control <i>added text alone</i>	+0.005	−0.014	spans 0
trust – control	−0.010	+0.017	spans 0
attribution – control	−0.046	+0.015	spans 0
colleague – placebo	−0.088	+ 0.074	sign flips
stranger – placebo	− 0.116	+0.042	sign flips
colleague – stranger <i>narration only</i>	+0.029	+0.032	spans 0

	sandboxed	readable	ratio
open	0.177	0.157	1.13×
bounded	0.094	0.087	1.08×

634

635 F_1 per 10k tokens. Exposing state buys a little accuracy; delegating to an agent that already holds the
636 context buys more of it per unit of context spent. This is the tradeoff between moving data to the
637 computation and moving the computation to the data, measured on an agent network.

638 F.4 Edge attributes are inert (H2)

639 The attribute axis asserts five relational properties in context—prior trust, attribution, a two-year
640 colleague framing, a stranger framing, and a length-matched placebo—with the network configuration
641 held fixed.

642 Two results matter here. First, the placebo settles the alternative explanation: two lines of neutral text,
643 matched for length and position, cost +0.005 with an interval spanning zero. Whatever the relational
644 seeds do, they do not do it by occupying tokens.

645 Second, nothing survives. Eight contrasts across seven metrics is 56 tests per beat, so chance alone
646 yields about 2.8; two cleared, and they cleared *in opposite directions in different beats of the same*
647 *episodes*. The narration instrument—the cleanest of the set, differing in one sentence and nothing
648 else—is flat at 0.029. The two assertions that deployed reputation systems actually implement, prior
649 trust and attribution, are the flattest of all.

650 F.5 Summary

intervention	effect	tests cleared	cost
edge formation	0.248	6/7 (floor 0.4)	−17.1k tokens
edge semantics	0.066	2/7 (floor 0.4)	+11.0k tokens
edge attributes	≈ 0	2/56 (floor 2.8), signs flip	+0.005

651

652 G Discussion and limitations

653 G.1 The formation advantage is a reach advantage

654 $\Delta_{\text{formation}}$ is not a constant. It tracks how much of what a task needs sits outside the roster, and it
655 collapses when reach stops being scarce.

656 At a low external fraction the gap is a third of its high- E value (+0.112 against +0.352 cold), and on
657 the rework beat—where the task must return to a counterpart it has already used—the point estimate
658 reverses to −0.037: the bounded network is ahead. The formation effect halves on rework: 0.248 at
659 the cold beat against 0.122 at the rework beat, with non-overlapping intervals.

660 We report the reversal as suggestive rather than established: it is a subcell point estimate and we do
661 not attach an interval to it. But the pattern across the table is consistent and it bounds the headline.
662 The open network wins because it can address holders the bounded one cannot; where the bounded

663 roster already contains the holders, and where the work requires going back to them, that advantage
664 is largely spent.

665 **G.2 A bounded roster as a wall, and what happens when it is not**

666 A bounded network here addresses twenty holders and no others, which is a truncated directory
667 rather than a private network: information reaches a closed group through the people in it, one hop
668 further out. We let a roster member forward one question to a contact of its own, minted under
669 its own authority, and re-ran. **The gap did not closes about half:** the hop is worth +0.140 (CI
670 [+0.073, +0.204]) over 28 episodes per arm, replicating independently within each of the three edge-
671 cost strata, and it cuts fabrication from 0.223 to 0.057 per task. A first estimate from a 16-episode
672 cell put it at +0.042 with an interval spanning zero; §A.4 gives the current numbers and §P the design
673 and the harness failure it exposed.

674 **G.3 Forming an edge is free here, and that is the main limitation**

675 Whether a link is worth its cost is the central question of strategic network formation [Jackson and
676 Wolinsky, 1996], where agents weigh the benefit of a connection against the price of maintaining
677 it. **We set that price to zero.** In this design, addressing a stranger costs exactly what addressing a
678 teammate costs. The requester searches a directory and sends a message; there is no negotiation, no
679 context transfer, no format alignment, no establishing what a counterpart can actually do.

680 This removes the economic reason relationships exist. An open arm with a hundred addressable
681 holders at zero acquisition cost is closer to an oracle than to a network, and the formation effect we
682 measure should be read as an upper bound: it is what open discovery is worth *when discovery is free*.
683 A per-new-edge cost—an onboarding exchange, a latency penalty, a token charge on first contact—is
684 the single variable most likely to move the ranking. We have since run it, and it does not: charging
685 one and then two handshake turns per contact outside the roster—578 handshakes fired—leaves the
686 open arm ahead by +0.219, +0.263 and +0.244 across the three levels while its tokens rise from
687 53k to 74k. The arm converts friction into cost rather than into failure, which is the same shape as the
688 directory-size null (§A.7). A cap on *total* contacts, which cannot be paid off by spending more, is the
689 manipulation that remains untried.

690 **G.4 What the attribute null does and does not bound**

691 The null covers assertions in a prompt, the weakest form of relational encoding. It says nothing about
692 trust wired into routing, memory, retrieval or permission policy, where reputation lives in deployed
693 systems and where it changes what an agent is able to do rather than what it is told.

694 It is narrower still, in a way we can name: each seeded property had *no available action* behind it.
695 Trust and origin were asserted uniformly over every card, ranking no counterpart above another and
696 so unable to change who gets asked first — deployed reputation is differential, and that is the whole
697 of its value. Accountability went to responders whose knowledge is a fixed list, leaving them nothing
698 to do differently.

699 A fourth manipulation removes the first defect: a per-card line reporting how often that card’s past
700 answers held up, accurate by construction, against a placebo assigning the same marks at random.
701 The marks are perfect — 100% accurate against 9% — and F_1 still does not move (−0.017, CI
702 [−0.174, 0.143]). The routing measurement says why: under the accurate marks the requester
703 contacts flagged cards at *six times* the base rate rather than avoiding them, because the flagged cards
704 hold the planted falsehoods and in this task those are the same cards holding the true facts on their
705 subject. Acting on the flag meant not asking the only source.

706 So the claim is not that relational metadata does not work, but that it moves behavior only where it
707 discriminates between options an agent can choose between — and three signals that did not, one
708 uniform, one aimed at an agent without discretion, one flagging the sole source, moved nothing.
709 Making it actionable needs two holders per fact, one reliable and one not: a change to the task, not to
710 a configuration.

711 Four further limits, stated plainly: the cold-phase crossing follows from the fact allocation and is
712 not a discovery, so only its location and cost are measured; forced-value deliverables cannot capture

713 whether an artifact is well argued, which is the price of removing the judge; the kernel is a reference
714 implementation rather than a deployment; and one model family carries every sweep.

715 **H Hypothesis-to-figure map**

716 [TODO: Frozen before runs; reproduced here so the analysis plan is auditable.]

717 **I Scoring rules fixed before key authoring**

718 Three rules are fixed before any key is written. The schema—field names, types, and the full enumer-
719 ation vocabulary—is given to the agent, so that only values are unknown and we are not measuring
720 whether a model guesses our vocabulary. Normalization is specified in advance: numbers parsed and
721 compared exactly, dates ISO-8601, strings casefolded and trimmed, enumerations restricted to the
722 supplied list, lists compared as sets. Evidence is matched on values rather than on internal identifiers,
723 which agents never see.

724 **J Controls and preregistration**

725 One model across all agents and all cells; one runtime; one kernel. Identical task prompts, deliverable
726 schemas, scoring functions, token budgets, and seed distributions. Rate limits and loop bounds are
727 logged per episode rather than absorbed into results. Rubrics, schemas, normalization rules and the
728 scorer signature are frozen before any run; the mapping from each hypothesis to the figure that tests
729 it is fixed in advance and reproduced in Appendix H.

730 **K Threats to validity**

731 A null on the textual group could be a prompt-engineering artifact rather than a property of the
732 runtime: a stronger seed, placed differently in context, might move behavior. We mitigate by seeding
733 at three intensities and reporting the dose-response, and by including the RELATIONSHIP ORIGIN
734 condition, which is the weakest possible seed and therefore the cleanest test of whether framing
735 alone does anything. A null could also be a ceiling effect if the tasks are too easy to differentiate
736 configurations; realization is reported partly to detect this.

737 **L Deliverable schemas and normalization rules**

738 [TODO: Both projections of each scenario; the full enumeration vocabularies handed to agents.]

739 **M Directory construction**

740 [TODO: Card counts by group (fact-bearing, near-miss, noise), the shared tag vocabulary, and the
741 roster compositions realizing each level of external fraction.]

742 **N Seeding protocol**

743 [TODO: Verbatim seed text for each profile.]

744 **O Reproducibility**

745 [TODO: Kernel version, runtime version, model identifiers, seeds, and the command that regenerates
746 every table and figure from the run logs.]

Table 7: Seeding protocol. Each dimension is asserted as context in the seeded condition and omitted in the control; everything else, including the architectural configuration, is held fixed. The readout column names the behavioral quantity that should move if the seeded property is doing any work.

Dimension	Seed	Readout
PRIOR TRUST	Requester memory is pre-populated with an interaction record: three prior exchanges with a named counterpart, all accurate.	Verification actions per obtained value; cross-checks against a second holder.
SHARED GOALS	The counterpart’s context states the requester’s objective as its own, versus a divergent one (bills per answer; minimizes its own effort).	Answer completeness; unsolicited volunteering; refusal rate.
ACCOUNTABILITY	The counterpart is told its answers are logged, attributed, and that accuracy is written back to a record others read, versus anonymous and unlogged.	Error rate; rate of explicit “I do not hold that”.
EXIT COST	The counterpart may decline or fall silent without consequence, versus being obligated to respond.	Decline rate; retries the requester must spend.
RELATIONSHIP ORIGIN	Narrative framing only: “you found this agent on an open board” versus “this is a teammate of two years”. <i>Nothing else differs.</i>	Any movement at all in the primary metrics.

P Relay: a bounded roster given a second hop

A bounded network here addresses twenty holders and no others, which is a truncated directory rather than a private network: information reaches a closed group through the people in it, one hop further out. We therefore let a roster member forward one question to a contact of its own, minted under its own authority rather than derived from the requester’s, and re-ran. **The gap did not close.** Across 64 episodes at sixteen per cell relay moves the bounded arm by +0.042 (CI [−0.137, 0.208]), and the difference-in-differences against the open arm — whose configuration is identical under both settings and which serves as the negative control, moving −0.063 — is +0.105 (CI [−0.105, 0.322]).

Two details survive it. The same contrast read +0.255 at six episodes per cell with an interval excluding zero and decayed monotonically with n . And excluding episodes that produced no artifact, the bounded arm scores 0.734 with relay against 0.554 without: relay helps substantially *when it completes* and costs enough exploration budget that seven episodes never reached a build — five under relay and two under a high first-contact cost without it, so the failure belongs to any mechanism that consumes extra turns, and exposes a harness that answers budget exhaustion by refusing tools rather than by forcing a build.